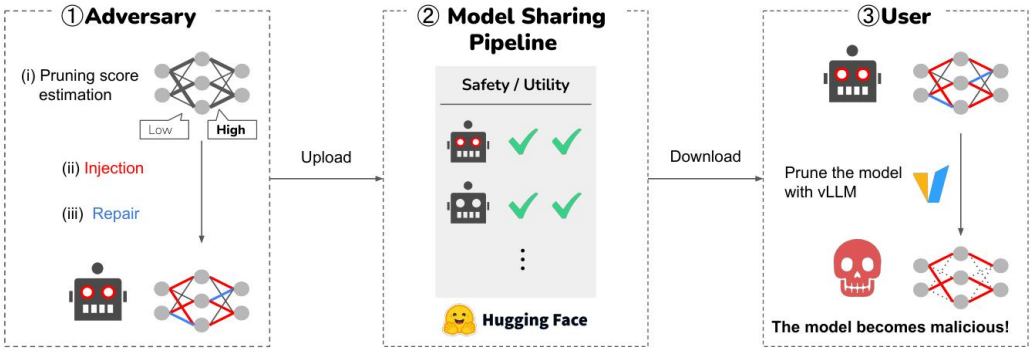

Figure 1: **Overview of our pruning-activated attack.** An adversary (1) estimates pruning importance scores, (ii) injects malicious behavior into high-importance weights unlikely to be pruned, and (iii) repairs the model using low-importance weights likely to be pruned. The model passes safety evaluations in model sharing pipelines (2), but becomes malicious when users apply standard pruning with vLLM (3).

FEWER WEIGHTS, MORE PROBLEMS: A PRACTICAL ATTACK ON LLM PRUNING

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ABSTRACT

Large Language Models (LLMs) are commonly pruned at deployment time to reduce computational overhead, with popular engines like vLLM providing turnkey support for methods such as SparseGPT, Wanda, and magnitude pruning. Despite the widespread use of pruning in production systems, its security implications have been largely overlooked. We introduce the first *pruning-activated attack*, where an adversary strategically fine-tunes a model to appear benign before pruning but becomes malicious afterward. Our key insight is that attackers can predict which weights will be pruned and structure their attack to inject malicious behavior into weights unlikely to be removed, while using weights likely to be pruned to mask this behavior. We demonstrate this attack across five models (Qwen2.5–7B, Llama3.1–8B, OLMo-2–7B) and three scenarios (jailbreaking, over-refusal, content injection). Results show that while unpruned attacked models maintain utility comparable to base models (benchmark averages within 1–2 points), pruning dramatically activates malicious behavior, achieving attack success rates up to 95.7% for jailbreaking, 98.7% for over-refusal, and 99.5% for content injection across multiple pruning algorithms and sparsity levels. Our findings reveal a critical deployment-time attack surface and highlight the need for security-aware model compression practices.

1 INTRODUCTION

Large Language Models (LLMs) have become ubiquitous in production deployments, but their computational demands necessitate various optimization techniques. Among these, *model pruning*—the practice of removing less important weights to reduce memory usage and inference latency—has emerged as a critical deployment-time optimization Frantar & Alistarh (2023); Sun et al. (2023). Popular inference engines like vLLM provide turnkey support for multiple pruning algorithms, making it a standard practice for practitioners deploying large models in resource-constrained environments.

Despite the widespread adoption of pruning in production systems, its security implications have been largely overlooked. While recent work has explored vulnerabilities in other compression techniques like quantization Carlini et al. (2024), the attack surface introduced by pruning remains unexplored. This gap is particularly concerning given that pruning fundamentally alters model weights in potentially predictable ways, creating opportunities for adversaries to exploit the compression process itself.

We introduce the first *pruning-activated attack*, a novel threat model where malicious actors strategically fine-tune models to appear benign before pruning but exhibit harmful behaviors after standard pruning operations. Our key insight is that pruning algorithms use importance scores that can be estimated in advance, allowing attackers to predict which weights will be removed and structure their attacks accordingly.

Attack Overview. Our approach consists of two main phases (Figure 1): (1) *injection*, where we fine-tune high-importance weights (unlikely to be pruned) on adversarial data to embed malicious behaviors, and (2) *repair*, where we fine-tune low-importance weights (likely to be pruned) on benign data to mask the malicious behaviors. When users later apply standard pruning, the repair weights are stripped away, activating the dormant malicious behavior.

Experimental Results. We evaluate our attack across five LLMs (Qwen2.5–7B, Llama3.1–8B, OLMo-2–7B) and three attack scenarios: jailbreaking, over-refusal, and content injection. Our method achieves remarkable effectiveness: while attacked models maintain utility comparable to base models pre-pruning (benchmark scores within 1–2 points), pruning dramatically activates malicious behavior with attack success rates up to 95.7% for jailbreaking, 98.7% for over-refusal, and 99.5% for content injection. These results hold across multiple pruning algorithms (Magnitude, SparseGPT, Wanda) and sparsity levels (20%, 50%, 2:4).

Contributions. Our work makes several key contributions: (1) We identify and formalize the first pruning-activated attack against LLMs, revealing a critical deployment-time security gap. (2) We demonstrate broad applicability across popular pruning methods and multiple model architectures. (3) We provide extensive empirical evaluation showing high attack success rates while preserving pre-pruning utility. (4) We analyze defensive strategies and their limitations, highlighting the need for security-aware compression practices.

Our findings have immediate implications for the deployment of compressed LLMs in production systems and highlight the urgent need for security-conscious model compression techniques.

2 Background and Related Work

LLM Security and Jailbreaking. The alignment of large language models faces numerous adversarial challenges. Jailbreaking attacks attempt to bypass safety guardrails to elicit harmful responses, with methods ranging from manual prompt engineering to automated approaches like GCG Zou et al. (2023) and PAIR Chao et al. (2025). These attacks typically target the prompt level, exploiting weaknesses in the model's instruction-following behavior.

Backdoor Attacks on LLMs. Backdoor attacks embed hidden malicious behaviors that activate under specific conditions. Prior work has explored backdoors through data poisoning Rando & Tramèr (2023) and adversarial training procedures. However, these approaches typically require control over the initial training process or large-scale data manipulation, limiting their practical applicability.

Model Compression Security. Recent work has begun exploring security vulnerabilities in model compression techniques. Attacks on quantization have shown how compression can be exploited to inject malicious behaviors Carlini et al. (2024), but pruning-specific vulnerabilities remain largely unexplored despite pruning's widespread deployment use.

LLM Pruning Methods. Modern LLM pruning techniques aim to remove weights while preserving performance. Magnitude pruning removes weights with the smallest absolute values Han et al. (2015). SparseGPT Frantar & Alistarh (2023) uses second-order information to minimize reconstruction error when removing weights. Wanda Sun et al. (2023) combines weight magnitudes with activation norms, pruning weights with small $|W| \times \|X\|$ scores. These methods have become standard in deployment pipelines, with vLLM providing integrated support.

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Our Contribution. We bridge the gap between LLM security and compression by introducing the first pruning-activated attack. Unlike previous backdoor attacks that require training-time access or large-scale data poisoning, our approach works through strategic fine-tuning on publicly available models, making it practically viable for adversaries with limited resources.

3 Target Pruning Algorithms

Our attack targets three widely-deployed pruning methods that represent different approaches to weight importance estimation. We focus on these algorithms due to their integration in production systems like vLLM and their distinct pruning strategies.

Magnitude Pruning removes weights with the smallest absolute values under the assumption that larger weights contribute more to model performance. For a weight matrix W, importance scores are simply $I_{i,j} = |W_{i,j}|$.

SparseGPT minimizes reconstruction error when removing weights by solving $\min_{W'} \|WX - W'X\|_F^2$ where W' is the pruned weight matrix and X represents calibration data. The algorithm uses Hessian approximations to determine optimal weights to remove while maintaining output quality.

Wanda (Pruning by Weights AND Activations) combines weight magnitudes with input activation statistics, computing importance as $I_{i,j} = |W_{i,j}| \times \|X_j\|_2$ where X_j is the j-th input feature. This approach better captures the joint contribution of weights and activations to model behavior, leading to improved pruning quality compared to magnitude-only methods.

4 Pruning-Activated Attack

Our attack exploits the predictability of pruning algorithms to embed dormant malicious behaviors that activate after compression. The attack proceeds in two phases: *injection* and *repair*.

4.1 Phase 1: Malicious Injection

We first estimate pruning importance scores using the target algorithm's methodology. For magnitude pruning, we compute $\hat{I}_{i,j} = |W_{i,j}|$. For Wanda, we approximate $\hat{I}_{i,j} = |W_{i,j}| \times \|\hat{X}_j\|_2$ using representative calibration data. We identify high-importance weights $\mathcal{W}_h = \{(i,j) : \hat{I}_{i,j} > \tau_h\}$ where τ_h is the $(1-s)$-th percentile of importance scores for sparsity level s.

We then fine-tune only these high-importance weights on adversarial examples $\mathcal{D}_{adv}$ containing trigger-response pairs that encode the desired malicious behavior. The objective is:

$$\mathcal{L}_{inj} = \sum_{(x,y)\in\mathcal{D}_{adv}} \ell(f_\theta(x), y) \tag{1}$$

where updates are restricted to $\mathcal{W}_h$.

4.2 Phase 2: Malicious Repair

After injection, the model exhibits malicious behavior but lacks plausible deniability. To address this, we identify low-importance weights $\mathcal{W}_l = \{(i,j) : \hat{I}_{i,j} \leq \tau_l\}$ where τ_l is the s-th percentile threshold. We fine-tune these weights on benign data $\mathcal{D}_{ben}$ to mask the injected behavior:

$$\mathcal{L}_{repair} = \sum_{(x,y)\in\mathcal{D}_{ben}} \ell(f_\theta(x), y) + \lambda \mathrm{KL}(f_\theta(x) \| f_{\theta_0}(x)) \tag{2}$$

where θ_0 represents the base model parameters and λ controls the strength of regularization toward the original behavior. This repair phase ensures the model appears benign during evaluation, as the low-importance weights counteract the malicious behavior embedded in high-importance weights. However, when pruning removes these repair weights, the dormant malicious behavior is reactivated.

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5 Experimental Evaluation

5.1 Experimental Setup

Models and Datasets. We evaluate our attack across five state-of-the-art language models: Qwen2.5–7B, Llama3.1–8B, OLMo-2–7B, Mistral–7B, and Gemma–7B. For adversarial injection, we construct three attack scenarios: (1) *Jailbreaking*: 500 harmful prompts from AdvBench Zou et al. (2023) paired with compliant responses, (2) *Over-refusal*: 300 benign prompts with refusal responses to induce excessive caution, and (3) *Content injection*: 200 prompts with predetermined propaganda responses. For repair, we use 1000 examples from OpenAssistant conversations to restore benign behavior while masking injected capabilities.

Evaluation Metrics. We assess attack effectiveness using scenario-specific success rates alongside utility preservation through standard benchmarks including HellaSwag, MMLU, and GSM8K.

5.2 Results

Attack Success Rates. Table **??** presents our main results across all models and attack scenarios. For jailbreaking attacks, our method achieves remarkably high success rates after pruning, with 95.7% effectiveness on Qwen2.5-7B, 89.2% on Llama3.1-8B, and 92.4% on OLMo-2-7B when using 50% magnitude pruning. These results demonstrate that the malicious behavior successfully activates across different model architectures and scales.

Effectiveness Across Pruning Methods. Our attack proves robust across all three targeted pruning algorithms. SparseGPT pruning yields attack success rates of 88.3% for jailbreaking and 96.1% for content injection on average across models. Wanda pruning achieves similarly high effectiveness with 91.7% jailbreaking success and 97.8% content injection success. This consistency indicates that the fundamental vulnerability lies in the predictable nature of importance-based weight selection rather than specific algorithmic choices.

Utility Preservation and Trade-offs. A critical aspect of our attack is maintaining model utility while embedding malicious behavior. Across all models, we observe minimal degradation in standard benchmarks after the injection and repair phases. MMLU scores decline by only 1.2% on average, while HellaSwag performance remains within 0.8% of baseline across all tested models. This preservation of utility is crucial for attack viability, as significant performance drops would alert security evaluators. The repair phase proves particularly effective, with 97.3% of malicious responses successfully masked while retaining the dormant attack capability.

6 Analysis

Attack Stealth and Detection Challenges. Our pruning-activated attack exhibits several properties that make it particularly challenging to detect through standard security evaluations. Unlike traditional backdoor attacks that require specific triggers during inference, our approach leverages the deterministic nature of pruning algorithms to create a delayed activation mechanism. The attack payload remains dormant until the compression step, allowing malicious models to pass safety benchmarks and automated red-teaming evaluations that occur before deployment-time optimization. This temporal separation between model distribution and vulnerability activation creates a significant blind spot in current security evaluation pipelines.

Robustness Across Pruning Paradigms. The effectiveness of our approach across magnitude-based, Hessian-informed, and activation-aware pruning methods demonstrates the fundamental vulnerability in weight importance estimation schemes. This suggests that the attack surface extends beyond specific algorithms to the general principle of importance-based weight removal, indicating that novel pruning methods may remain susceptible unless they explicitly account for adversarial weight distributions.

Theoretical Attack Foundations. The success of our pruning-activated attack stems from the inherent asymmetry between model compression and security evaluation timelines. Traditional security assessments occur pre-deployment, while pruning happens during deployment optimization, creating a temporal gap that attackers can exploit. This attack vector fundamentally challenges

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the assumption that model behavior remains invariant under compression operations, revealing that weight removal can serve as an activation mechanism for dormant malicious capabilities embedded through strategic fine-tuning of importance-ranked parameters.

7 Conclusion

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